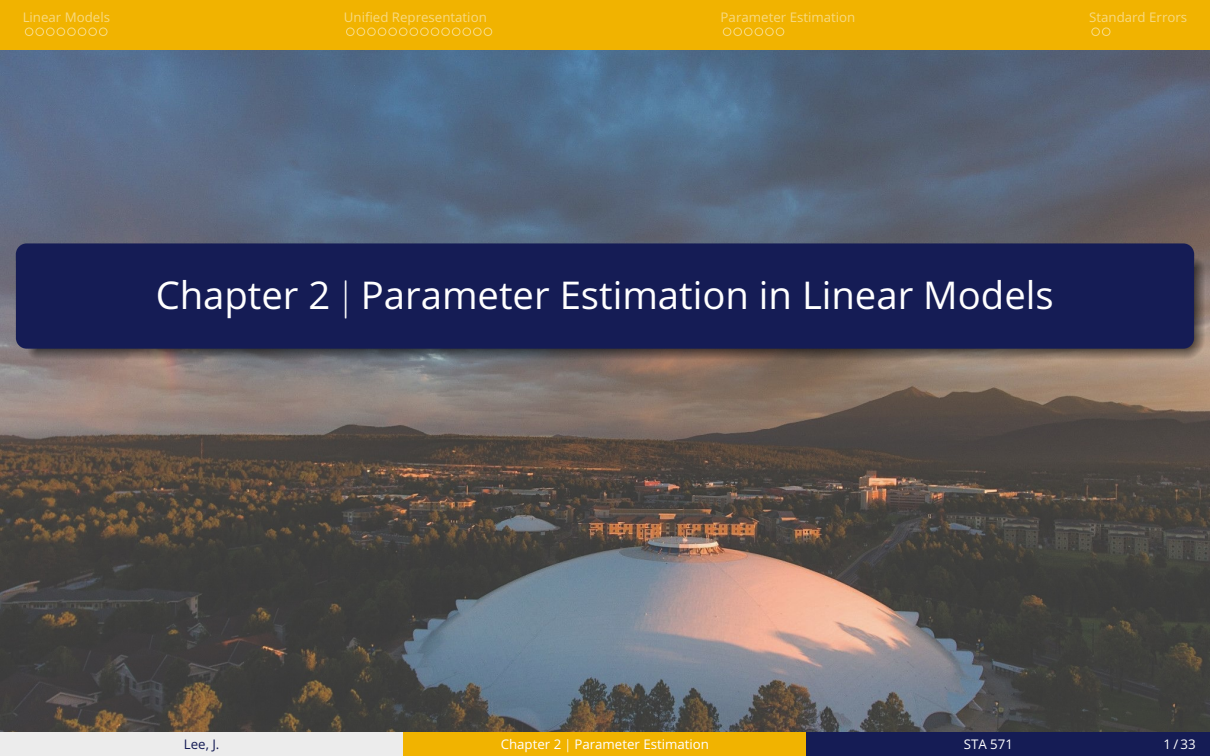


Chapter 2 | Parameter Estimation in Linear Models



Learning objectives

- Write simple regression models and one-way ANOVA models in a unified linear model representation
 - ① $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$, or
 - ② $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$
- Understand the principles of the least squares (LS) estimation, and use R programming to compute LS estimates for the regression coefficients $\boldsymbol{\beta}$ and model error variance σ^2 :
 - ① $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
 - ② $\hat{\sigma}^2 = \frac{1}{n-p^*} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n-p^*} (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}})$, where $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ and p^* denotes the number of the model coefficients in $\boldsymbol{\beta}$.
- Understand the uncertainty of the least squares (LS) estimation, and compute the standard errors of LS estimates using R programming.

Outline

- 1 Linear Models
- 2 A Unified Representation of Linear Models
- 3 Parameter Estimation in Linear Models
- 4 Standard Errors for Parameter Estimates

Linear Models (LMs)

- A **linear model (LM)** is a model where the data (predictors $\mathbf{x}_1, \dots, \mathbf{x}_p$ and response $\mathbf{y}$) and the parameters of interest $\beta = (\beta_1, \dots, \beta_p)^T$ interact only via addition and multiplication.
- Examples of linear models:
 - ① Simple linear regression: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$
 - ② Quadratic (polynomial) regression: $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i$
 - ③ Multiple regression: $y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i$
- Note: Meaning of “linear” in linear models – **linear in terms of the parameters of interest**
 - ① $y_i = \beta_0 e^{-\beta_1 x_i} + \epsilon_i$: non-linear model
 - ② $y_i = \beta_0 + \beta_1 \ln x_i + \epsilon_i$: linear model

LM example | Simple linear regression (SLR)

- A regression model is essentially a model of the relationships between **predictors** (or **covariates**) and a **response variable** (or **outcome**).
 - ① Similar to correlation in that we examine the linear relationship between two variables X & Y .
 - ② However, regression distinguishes between the response variable Y and the predictor X .
 - ③ The goal is to predict Y from X , or explain how Y changes with X using a straight-line relationship.
- A regression involves a model for how Y changes with X , which is based on the equation of a straight line: $Y = \text{intercept} + \text{slope} \cdot X$. Because Y cannot be expected to lie exactly on a line with X , the model adds error:

$$Y = \text{intercept} + \text{slope} \cdot X + \text{error}$$
$$\Rightarrow Y = \beta_0 + \beta_1 X + \epsilon$$

- **Interpretation:** for every unit increase in X (every time we add 1 to X), we estimate the **mean (or expected value) of Y** (the mean response) to increase by β_1 units.

LM example | Inference for SLR

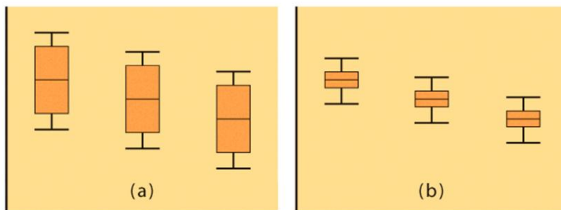
- Statistical inference for SLR $Y = \beta_0 + \beta_1 X + \epsilon$
 - ① The intercept β_0 and slope β_1 of this relationship have to be estimated from sample data on X and Y .
 - ② The preferred method of doing this estimation is called the method of least squares, and produces the straight line that goes through the points in a scatterplot of Y vs. X which is closest (in some sense) to all of the points.
 - ③ We are often interested in the point estimate of β_1 . This tells how much Y changes (on average) for a given one-unit increase in X .
 - ④ We are also often interested in testing $H_0 : \beta_1 = 0$ (whether the slope is 0). This test gets at whether or not Y depends (linearly) on X . That is, where there is a linear trend in Y as X increases.

LM example | One-way analysis of variance (ANOVA)

- A **factor** is a variable that can take one of several levels used to differentiate one group from another.
- An experiment has a **one-way design (completely randomized design)** if several levels of one factor are being studied and the individuals are randomly assigned to its levels.
 - ① Example: Four levels of nematode quantity in seedling growth experiment. This is a one-way design, because there is only one way to group the data.
 - ② However, two seed species and four levels of nematodes would be a two-way design.

LM example | One-way ANOVA

- **Analysis of Variance (ANOVA)** is a technique used to determine whether more than two population means are equal or at least one of them is different from the others.
 - 1 One-way ANOVA is used for one-way designs (completely randomized designs).
 - 2 We want to know if the observed differences in sample means are likely to have occurred by chance just because of random sampling.



- 3 This will likely depend on (a) the difference between the sample means and (b) how much variability there is within each sample

LM example | One-way ANOVA

- Random sampling always produces chance variations. Any “factor effect” would thus show up in our data as the factor-driven differences plus chance variations (“error”):

$$\text{Response} = \text{fit ("factor" or "groups")} + \text{residual ("error").}$$

- One-way ANOVA setting:
 - ① We have m independent simple random samples (SRS) from m populations or treatments or groups.
 - ② The i -th population has a normal distribution with unknown mean μ_i .
 - ③ All m populations have the identical unknown variance σ^2 .

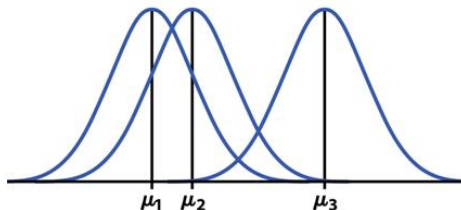
LM example | One-way ANOVA

- The **one-way ANOVA model** analyses situations where chance variations ϵ 's are normally distributed $N(0, \sigma^2)$ so that:

$$Y_{ij} = \mu_i + \epsilon_{ij}$$

for $i = 1, \dots, m$ and $j = 1, \dots, n_i$. (We denote $n = \sum_{i=1}^m n_i$ for later use.)

- 1 The error (chance variation) ϵ_{ij} 's are assumed to follow an $N(0, \sigma^2)$ distribution.
- 2 The unknown parameters are the population means $\mu_1, \dots, \mu_m$ and the common variance σ^2 .

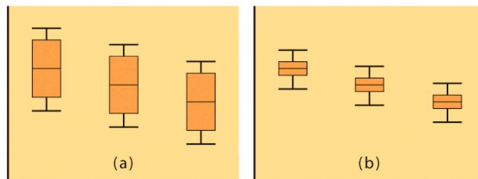


- Test hypotheses:** $H_0 : \mu_1 = \mu_2 = \dots = \mu_m$ vs. $H_a : \text{not all } \mu_i\text{'s are equal}$

LM example | One-way ANOVA

- The ANOVA F -statistic compares variation due to specific sources (levels of the factor) with variation among individuals who should be similar (individuals in the same sample):

$$F = \frac{\text{variation among sample means}}{\text{variation among individuals in same sample}}$$



- The difference in means is **small** relative to within-sample variability $\Rightarrow F$ tends to be **small** $\Rightarrow$ **Do not reject H_0**
- The difference in means is **large** relative to within-sample variability $\Rightarrow F$ tends to be **large** $\Rightarrow$ **Reject H_0**

A unified representation of linear models

- Linear models have the unified matrix representation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

- 1 $\mathbf{y} = (y_1, \dots, y_n)^T$ is a vector of n **observable** responses.
- 2 $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_p)$ is an $n \times p$ **design (model) matrix** with j th **observed** predictor $\mathbf{x}_j = (x_{1j}, \dots, x_{nj})^T$ for $j = 1, \dots, p$.
- 3 $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$ is a p -dim vector of **unknown** regression parameters.
- 4 $\boldsymbol{\varepsilon} = (\epsilon_1, \dots, \epsilon_n)^T$ is an n -dim vector of **unobservable** model errors with mean $\mathbf{0}$ and variance-covariance matrix $\boldsymbol{\Sigma}$. That is,

$$E(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \boldsymbol{\Sigma}$$

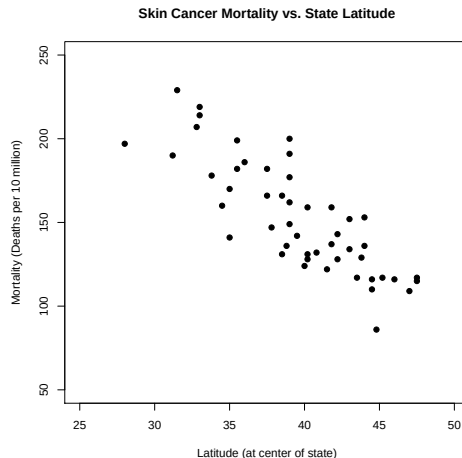
- 5 If the model errors $\epsilon_1, \dots, \epsilon_n$ are **independent and identically distributed** with **common** variance σ^2 , then

$$E(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$$

SLR | Data example

- **Example 2.1:** The response variable Y is the mortality due to skin cancer (number of deaths per 10 million people) and the predictor variable X is the latitude (degrees North) at the center of each of 49 states in the U.S.¹

	State	Lat	Mort	Ocean	Long
1	Alabama	33.0	219	1	87.0
2	Arizona	34.5	160	0	112.0
3	Arkansas	35.0	170	0	92.5
4	California	37.5	182	1	119.5
5	Colorado	39.0	149	0	105.5
6	Connecticut	41.8	159	1	72.8
7	Delaware	39.0	200	1	75.5
8	Wash,D.C.	39.0	177	0	77.0
:	:	:	:	:	:
45	Virginia	37.5	166	1	78.5
46	Washington	47.5	117	1	121.0
47	WestVirginia	38.8	136	0	80.8
48	Wisconsin	44.5	110	0	90.2
49	Wyoming	43.0	134	0	107.5



¹ Dataset: `skin.cancer.txt`. The data were compiled in the 1950s, so Alaska and Hawaii were not yet states. And, Washington, D.C. is included even though it is not technically a state.

SLR | Model formulation

- Our goal is to predict the mortality Mort of a state, based on the latitude Lat of the state.

① Mort is the response (outcome variable); the predictor (explanatory variable) is Lat .

② We assume

$$\text{Mort} = \text{intercept} + \text{slope} \times \text{Lat} + \text{error}$$

③ A regression model is a model of the **average response at a given value of the predictor**. In this example, the regression model is a model of the following conditional expectation:

$$E(\text{Mort}|\text{Lat}) = \text{intercept} + \text{slope} \times \text{Lat}$$

which is a linear function of Lat (and also a linear function of intercept and slope).

④ For a pair $(\text{Lat}_i, \text{Mort}_i)$ to observe from i th state, we assume that each Mort_i can be modeled by a linear function of the associated Lat_i plus an “error”:

$$\text{Mort}_i = \text{intercept} + \text{slope} \times \text{Lat}_i + \epsilon_i, \quad i = 1, \dots, n$$

⑤ If $E(\epsilon_i) = 0$, the **conditional mean of response Mort_i at a given value of Lat_i** from i th state is

$$E(\text{Mort}_i|\text{Lat}_i) = \text{intercept} + \text{slope} \times \text{Lat}_i, \quad i = 1, \dots, n$$

SLR | Model formulation

- In general, to predict a response Y based on a predictor X , SLR assumes

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Note: If $E(\epsilon) = 0$, the conditional expectation of Y given $X = x$ is

$$E(Y|x) = \beta_0 + \beta_1 x$$

- Now, for each pair (X_i, Y_i) to observe, SLR assumes that each Y_i can be modeled by a linear function of the associated X_i plus an “error”:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, n$$

Note: If $E(\epsilon_i) = 0$, the conditional expectation of Y_i given $X_i = x_i$ is

$$E(Y_i|x_i) = \beta_0 + \beta_1 x_i, \quad i = 1, \dots, n$$

SLR | Model formulation

- Why is it called a **simple linear regression** model?
 - 1 Because we are modeling the response Y on the predictor X alone, we only have one predictor: hence it is a **simple** model.
 - 2 In the regression model,

$$E(Y|x) = \beta_0 + \beta_1 x$$

The conditional expectation of Y given $X = x$ is linear in x (and also is linear in intercept β_0 and slope β_1 parameters). Therefore, it is a **linear** regression model.

SLR | Unified representation

- The SLR model can be expressed in a **unified representation**:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Why? We derive the unified representation of the SLR model as follows:

- The SLR model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ can be further specified for all $i = 1, \dots, n$:

$$\begin{aligned} \begin{aligned} Y_1 &= \beta_0 + \beta_1 X_1 + \epsilon_1 \\ &\vdots \\ Y_n &= \beta_0 + \beta_1 X_n + \epsilon_n \end{aligned} &\Rightarrow \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \beta_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_1 \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_1 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix} \end{aligned}$$

- Denote

$$\mathbf{y}_{n \times 1} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X}_{n \times 2} = \begin{bmatrix} 1 & X_1 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}, \quad \boldsymbol{\beta}_{2 \times 1} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \boldsymbol{\epsilon}_{n \times 1} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Then, the SLR model can be reexpressed as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

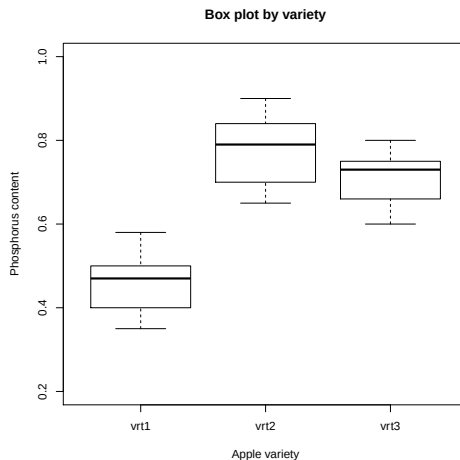
SLR | Research questions

- Most common problems of interest in SLR include:
 - ① Estimate the regression line $E(Y|x) = \beta_0 + \beta_1 x$ for the data. Interpret the line.
 - ② Calculate and interpret R^2 .
 - ③ Calculate and interpret the standard errors of the regression coefficient estimates.
 - ④ Interpret the estimated slope coefficient in terms of the variable `Mort` and `Lat`.
 - ⑤ Find and interpret a 95% confidence interval on the slope coefficient β_1 .
 - ⑥ Find and interpret a 95% prediction interval for the morality in Flagstaff, AZ.
 - ⑦ Are there any data points unusual compared to the others? If so, describe how.

One-way ANOVA | Data example

- **Example 2.2:** A horticulturist was investigating the phosphorus content of tree leaves from three different varieties of apple trees (`vrt1`, `vrt2`, `vrt3`). Random samples of five leaves from each of the three varieties were analyzed for phosphorus content.

	phos	variety
1	0.35	vrt1
2	0.40	vrt1
3	0.58	vrt1
4	0.50	vrt1
5	0.47	vrt1
6	0.65	vrt2
7	0.70	vrt2
8	0.90	vrt2
9	0.84	vrt2
10	0.79	vrt2
11	0.60	vrt3
12	0.80	vrt3
13	0.75	vrt3
14	0.73	vrt3
15	0.66	vrt3



One-way ANOVA | Model formulation

- **Cell-mean specification:** We start with a cell-mean specification of one-way ANOVA

$$y_{ij} = \mu_i + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

- 1 y_{ij} denotes the response from j th subject in i th group (or treatment).
 - 2 μ_i represents the **unknown** mean response of i th group (or treatment).
- **Offset specification:** An offset specification of one-way ANOVA model is

$$y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

- 1 μ denotes the **unknown** mean response of 1st group (treatment) as a baseline.
 - 2 τ_i is the difference of the **unknown** mean effect of i th group (treatment) from 1st group (treatment), with $\tau_1 = 0$ in R.
 - 3 $\mu_2 = \mu + \tau_2$, or $\tau_2 = \mu_2 - \mu_1$.
- Assumptions:
 - 1 The errors ϵ_{ij} 's are **independent and normally distributed** with **common** variance $\text{Var}(\epsilon_{ij}) = \sigma^2$.
 - 2 Equivalently, the responses y_{ij} 's are **independent and normally distributed**.

One-way ANOVA | Unified representation

- The one-way ANOVA cell-mean specification can be reexpressed as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Consider a cell-mean specification with $m = 3$ and $n_i = 5$.

- The one-way ANOVA model $Y_{ij} = \mu_i + \epsilon_{ij}$ can be further specified:

$$\begin{array}{l}
 Y_{11} = \mu_1 + \epsilon_{11} \\
 \vdots \\
 Y_{15} = \mu_1 + \epsilon_{15} \\
 Y_{21} = \mu_2 + \epsilon_{21} \\
 \vdots \\
 Y_{25} = \mu_2 + \epsilon_{25} \\
 Y_{31} = \mu_3 + \epsilon_{31} \\
 \vdots \\
 Y_{35} = \mu_3 + \epsilon_{35}
 \end{array}
 \Rightarrow
 \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \\ Y_{31} \\ \vdots \\ Y_{35} \end{bmatrix}
 = \mu_1 \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}
 + \mu_2 \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}
 + \mu_3 \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix}
 + \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \\ \epsilon_{31} \\ \vdots \\ \epsilon_{35} \end{bmatrix}
 \Rightarrow
 \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \\ Y_{31} \\ \vdots \\ Y_{35} \end{bmatrix}
 = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix}
 \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}
 + \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \\ \epsilon_{31} \\ \vdots \\ \epsilon_{35} \end{bmatrix}$$

One-way ANOVA | Unified representation

- (Continued)

② Denote

$$\underset{15 \times 1}{\mathbf{y}} = \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \\ Y_{31} \\ \vdots \\ Y_{35} \end{bmatrix}, \quad \underset{15 \times 3}{\mathbf{X}} = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix}, \quad \underset{3 \times 1}{\boldsymbol{\beta}} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}, \quad \underset{15 \times 1}{\boldsymbol{\epsilon}} = \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \\ \epsilon_{31} \\ \vdots \\ \epsilon_{35} \end{bmatrix}$$

Then, the one-way ANOVA model is expressed as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

One-way ANOVA | Unified representation

- The one-way ANOVA offset specification can also be reexpressed as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Consider an offset specification model with $m = 3$ and $n_i = 5$.

- The one-way ANOVA model $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$ can be further specified:

$$\begin{aligned}
 & \begin{aligned}
 Y_{11} &= \mu + \epsilon_{11} \\
 &\vdots \\
 Y_{15} &= \mu + \epsilon_{15} \\
 Y_{21} &= \mu + \tau_2 + \epsilon_{21} \\
 &\vdots \\
 Y_{25} &= \mu + \tau_2 + \epsilon_{25} \\
 Y_{31} &= \mu + \tau_3 + \epsilon_{31} \\
 &\vdots \\
 Y_{35} &= \mu + \tau_3 + \epsilon_{35}
 \end{aligned}
 \Rightarrow
 \begin{aligned}
 \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \\ Y_{31} \\ \vdots \\ Y_{35} \end{bmatrix}
 &= \mu \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}
 + \tau_2 \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}
 + \tau_3 \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix}
 + \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \\ \epsilon_{31} \\ \vdots \\ \epsilon_{35} \end{bmatrix}
 \end{aligned}
 \Rightarrow
 \begin{aligned}
 \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \\ Y_{31} \\ \vdots \\ Y_{35} \end{bmatrix}
 &= \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \end{bmatrix}
 \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix}
 + \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \\ \epsilon_{31} \\ \vdots \\ \epsilon_{35} \end{bmatrix}
 \end{aligned}
 \end{aligned}$$

One-way ANOVA | Unified representation

- (Continued)

2 Denote

$$\underset{15 \times 1}{\mathbf{y}} = \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{15} \\ Y_{21} \\ \vdots \\ Y_{25} \\ Y_{31} \\ \vdots \\ Y_{35} \end{bmatrix}, \quad \underset{15 \times 3}{\mathbf{X}} = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \end{bmatrix}, \quad \underset{3 \times 1}{\boldsymbol{\beta}} = \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix}, \quad \underset{15 \times 1}{\boldsymbol{\varepsilon}} = \begin{bmatrix} \epsilon_{11} \\ \vdots \\ \epsilon_{15} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{25} \\ \epsilon_{31} \\ \vdots \\ \epsilon_{35} \end{bmatrix}$$

Then, the one-way ANOVA model is expressed as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Unified representation | SLR vs. one-way ANOVA

- Comparison of the representation $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \varepsilon$ for SLR vs. ANOVA (cell-mean and offset)

① The major difference is in the **model (design) matrix**. That is,

$$\mathbf{X}_{n \times 2} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_i \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \text{ for SLR; } \mathbf{X}_{15 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix} \text{ for cell-mean ANOVA; } \mathbf{X}_{15 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \end{bmatrix} \text{ for offset ANOVA}$$

② Another difference is in the parameter representation:

$$\boldsymbol{\beta}_{2 \times 1} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \text{ for SLR; } \boldsymbol{\beta}_{3 \times 1} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} \text{ for cell-mean ANOVA; } \boldsymbol{\beta}_{3 \times 1} = \begin{bmatrix} \mu \\ \tau_2 \\ \tau_3 \end{bmatrix} \text{ for offset ANOVA}$$

Estimation of β | Least squares (LS) method

- For each pair (X_i, Y_i) to observe, SLR model assumes

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad \text{for } i = 1, \dots, n, \quad \text{or} \quad \mathbf{y} = \mathbf{X}\beta + \epsilon$$

How do we estimate the regression coefficients (β_0, β_1) (or $\beta = [\beta_0, \beta_1]^T$)?

- Least squares (LS) method** finds the values of (β_0, β_1) that minimize the sum of squared errors (SSE):

$$\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1]^T = \min_{\beta_0, \beta_1} \text{SSE}(\beta_0, \beta_1) = \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = \min_{\beta} (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$$

- LS estimates $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1]^T$ are then expressed as

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

or in matrix form,

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Estimation of β | LS method

- Notes on LS method

- With observed (x_i, y_i) , $SSE(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$ is a quadratic function of (β_0, β_1) .
- The LS estimates $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1]^T$ are then the solution of the following differential equations:

$$\begin{aligned} \frac{\partial SSE(\beta_0, \beta_1)}{\partial \beta_0} = 0 & \Rightarrow -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial SSE(\beta_0, \beta_1)}{\partial \beta_1} = 0 & \Rightarrow -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \end{aligned}$$

- LS estimates $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1]^T$ are linear combinations of the responses $y_1, \dots, y_n$, because

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \sum_{i=1}^n \left(\frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) y_i, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n \left(\frac{(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) y_i$$

Also, $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is linear in $\mathbf{y}$.

Residuals

- In SLR, the fitted value (predicted response) $\hat{y}_i$ is $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ for $i = 1, \dots, n$, or

$$\begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} \Rightarrow \hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

- ① Because $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$, we express $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H}\mathbf{y}$.
- ② $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ is called the **hat matrix**.
- The residual r_i is calculated by $r_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$ for $i = 1, \dots, n$, or

$$\begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{bmatrix} \Rightarrow \mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$

Residuals | Properties

- Linear projection theorem says that the vector of residuals $\mathbf{r}$ is the projection of $\mathbf{y}$ onto the orthogonal complement $L^\perp$ of the plane L spanned by $\mathbf{1} = [1, \dots, 1]^T$ and $\mathbf{x} = [x_1, \dots, x_n]^T$.
 - $\sum_{i=1}^n r_i = \mathbf{r}^T \mathbf{1} = 0$, meaning the residuals total is always equal to 0.
 - $\sum_{i=1}^n r_i x_i = \mathbf{r}^T \mathbf{x} = 0$, meaning $\mathbf{r}$ is linearly independent² of $\mathbf{x}$.
 - $\sum_{i=1}^n r_i \hat{y}_i = \mathbf{r}^T \hat{\mathbf{y}} = 0$, meaning $\mathbf{r}$ is linearly independent of $\hat{\mathbf{y}}$.
- In practices, these theoretical results are applied to model diagnostics. That is, if the fitted model is a “good” model, then
 - the plot of (x_i, r_i) 's should show no or very weak correlation.
 - the plot of $(\hat{y}_i, r_i)$'s should show no or very weak correlation.

²Two vectors $\mathbf{a}$ and $\mathbf{b}$ are called linearly independent or orthogonal, if the inner product $\mathbf{a}^T \mathbf{b} = 0$. Consider $\mathbf{a} = [1, 0]^T$ and $\mathbf{b} = [0, 2]^T$. They are orthogonal with $\mathbf{a}^T \mathbf{b} = 1 \cdot 0 + 0 \cdot 2 = 0$, indicating the angle between $\mathbf{a}$ and $\mathbf{b}$ is 90° .

Estimation of σ^2

- Now, we assume $\epsilon_1, \dots, \epsilon_n$ are IID $N(0, \sigma^2)$ or $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ for SLR models.
- If we **knew** (β_0, β_1) , then $\epsilon_i = y_i - \beta_0 - \beta_1 x_i$ and, because $\frac{\epsilon_i}{\sigma} \sim N(0, 1)$,

$$\sum_{i=1}^n \left(\frac{\epsilon_i}{\sigma} \right)^2 = \frac{\sum_{i=1}^n \epsilon_i^2}{\sigma^2} = \frac{SSE(\beta_0, \beta_1)}{\sigma^2} \sim \chi^2(n),$$

so that we obtain

$$E \left(\frac{\sum_{i=1}^n \epsilon_i^2}{\sigma^2} \right) = E \left(\frac{SSE(\beta_0, \beta_1)}{\sigma^2} \right) = n \Rightarrow E \left(\frac{\sum_{i=1}^n \epsilon_i^2}{n} \right) = \sigma^2$$

Therefore, $\frac{\sum_{i=1}^n \epsilon_i^2}{n}$ would be an unbiased estimate of σ^2 .

Estimation of σ^2

- However, in practices, (β_0, β_1) are **unknown**. We might think of using an “estimate” of ϵ_i instead:

$$\sum_{i=1}^n \left(\frac{r_i}{\sigma} \right)^2 = \frac{\sum_{i=1}^n r_i^2}{\sigma^2} = \frac{SSE(\hat{\beta}_0, \hat{\beta}_1)}{\sigma^2} \sim \chi^2(n-2),$$

so that we have

$$E \left(\frac{\sum_{i=1}^n r_i^2}{\sigma^2} \right) = E \left(\frac{SSE(\hat{\beta}_0, \hat{\beta}_1)}{\sigma^2} \right) = n-2 \Rightarrow E \left(\frac{\sum_{i=1}^n r_i^2}{n-2} \right) = \sigma^2$$

Therefore,

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n r_i^2}{n-2} = \frac{SSE(\hat{\beta}_0, \hat{\beta}_1)}{n-2} = MSE(\hat{\beta}_0, \hat{\beta}_1)$$

is an **unbiased estimate**³ of σ^2 .

³Note: Why $n - 2$? Because $\mathbf{r}$ is the projection of $\boldsymbol{\epsilon}$ onto an $n - 2$ dimensional subspace — hence we can write its norm as the sum of the squared $n - 2$ independent standard normal random variables.

Least squares (LS) estimators $\hat{\beta}$ | Properties

- Assume $\epsilon_1, \dots, \epsilon_n$ are IID $N(0, \sigma^2)$ or $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ for SLR models.
- Properties of LS estimators $\hat{\beta}$:
 - Because $\mathbf{y} = \mathbf{X}\beta + \epsilon$, $\mathbf{y}$ are normally distributed. That is, $\mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$
 - Because both $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)^T = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ and $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H}\mathbf{y}$ are linear transformations of $\mathbf{y}$, they are normally distributed.
 - $\hat{\beta}$ are unbiased estimators:

$$E(\hat{\beta}_0) = \beta_0, \quad E(\hat{\beta}_1) = \beta_1 \quad \Rightarrow \quad E(\hat{\beta}) = \beta$$

- $\hat{\beta}$ have variances

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right), \quad \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \Rightarrow \quad \text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

- These LS estimates variances are estimated by

$$\widehat{\text{Var}}(\hat{\beta}_0) = \hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right), \quad \widehat{\text{Var}}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \Rightarrow \quad \widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

Standard errors of $\hat{\beta}$

- The **standard error** of a parameter estimator is the **square root of the estimated variance of the estimator**.

- The standard errors of $\hat{\beta}$ are calculated by

$$\begin{aligned} SE(\hat{\beta}_0) &= \sqrt{\widehat{\text{Var}}(\hat{\beta}_0)} = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \Rightarrow SE(\hat{\beta}_0) = \hat{\sigma} \sqrt{(\mathbf{X}^T \mathbf{X})_{[1,1]}^{-1}} \\ SE(\hat{\beta}_1) &= \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)} = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \Rightarrow SE(\hat{\beta}_1) = \hat{\sigma} \sqrt{(\mathbf{X}^T \mathbf{X})_{[2,2]}^{-1}} \end{aligned}$$

where $(\mathbf{X}^T \mathbf{X})_{[i,i]}^{-1}$ denotes the i -th diagonal element of $(\mathbf{X}^T \mathbf{X})^{-1}$.

- The standard errors are used to compute confidence intervals. For example, a **95% confidence interval** for β_1 is

$$(\hat{\beta}_1 - 1.96SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96SE(\hat{\beta}_1)) = \left(\hat{\beta}_1 - \frac{1.96\hat{\sigma}}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}, \hat{\beta}_1 + \frac{1.96\hat{\sigma}}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \right)$$

if n is sufficiently large.